Integral periodic points of the Fibonacci trace map

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Abstract

We classify all integral periodic points of the Fibonacci trace map $T(x,y,z) = (y,z,yz-x)$, without restricting the period or the invariant level $K = x^2 + y^2 + z^2 - xyz$. The periodic set consists exactly of two fixed points, one orbit of length three, and three explicitly parametrized families of orbits of lengths four, six, and twelve. The latter families are classical; the assertion here is that their integral members, together with the special orbits, are exhaustive. A maximal-coordinate argument reduces an arbitrary integral periodic sequence to neighbours in $\{-1,0,1\}$, except for a completely resolved equality case at height two. Treating the zero and sign alternatives then leaves only the unit cube. It follows that every nonperiodic integral orbit leaves every bounded set in both time directions. On each integer level we give the exact fixed-point counts at every ordinary time and the resulting rational dynamical zeta function. Periodic points occur only on level zero, positive square levels, and levels $m^2 - m + 2$ with $m \geq 1$; the two infinite supports meet only at level four. The integral hypothesis and the distinction between one-map periodicity and whole-group finite orbits are essential.

1 The classification problem and main result

Consider the polynomial automorphism

$$T(x,y,z) = (y,z,yz-x), \qquad T^{-1}(x,y,z) = (xy-z,x,y). \tag{1.1}$$

Its integral dynamics preserve every level of

$$K(x,y,z) = x^2 + y^2 + z^2 - xyz, \qquad S_k(\mathbb{Z}) = \{P \in \mathbb{Z}^3 : K(P) = k\}, \quad k \in \mathbb{Z}. \tag{1.2}$$

The time variable throughout is the ordinary iterate of the single map T. In particular, we do not replace T by an iterate, identify points under coordinate sign changes, or ask for a finite orbit under every trace-map group element.

Periodic curves of this map over the real or complex numbers are well known. Their existence does not by itself determine the entire periodic set on the integer lattice: a classification must exclude all other periods, coordinate sizes, and levels at once. We give an elementary arithmetic argument that does so. The crucial use of integrality occurs after bounding the two neighbours of a coordinate of maximal modulus. There are only finitely many remaining local alternatives, and their resolution does not depend on the size of that maximum.

For $m \geq 1$, put

$$A_m = \mathrm{Orb}_T(m,0,0), \qquad B_m = \mathrm{Orb}_T(-1,m,-1), \qquad C_m = \mathrm{Orb}_T(1,m,1). \tag{1.3}$$

Here $\mathrm{Orb}_T(P) = \{T^j P : j \in \mathbb{Z}\}$. We also use

$$O = \{(0,0,0)\}, \quad E = \{(2,2,2)\}, \quad D = \{(-2,-2,2),(-2,2,-2),(2,-2,-2)\}. \tag{1.4}$$

For a finite orbit, its height means the largest absolute scalar coordinate at any point of that orbit.

Theorem 1.1 (Complete integral periodic locus). *The periodic set of* (1.1) *on* $\mathbb{Z}^3$ *is the disjoint union*

$$\mathrm{Per}(T, \mathbb{Z}^3) = O \sqcup E \sqcup D \sqcup \bigsqcup_{m \geq 1} (A_m \sqcup B_m \sqcup C_m). \tag{1.5}$$

The orbit data are as in Table 1. Consequently the set of least periods is exactly $\{1, 3, 4, 6, 12\}$.

An integral point has a bounded forward orbit, or a bounded backward orbit, if and only if it belongs to (1.5). *Every other integral point* P *satisfies*

$$\|T^n P\|_\infty \longrightarrow \infty \quad (n \to +\infty), \qquad \|T^n P\|_\infty \longrightarrow \infty \quad (n \to -\infty). \tag{1.6}$$

Orbit	Representative	Least period	Height	Level K
O	$(0, 0, 0)$	1	0	0
E	$(2, 2, 2)$	1	2	4
D	$(-2, -2, 2)$	3	2	4
A_m	$(m, 0, 0)$	6	m	m^2
B_m	$(-1, m, -1)$	4	m	$m^2 - m + 2$
C_m	$(1, m, 1)$	12	m	$m^2 - m + 2$

Table 1: Complete integral orbit types. The parameter in the last three rows ranges over all positive integers. Each row at each allowed parameter denotes one orbit, not just a collection of representatives.

The escape statement in the theorem is proper escape on a discrete lattice. It asserts neither monotonicity of the separate coordinates nor a quantitative growth rate. The classification also yields a sharp description on every level, even when $S_k(\mathbb{Z})$ itself is infinite.

Corollary 1.2 (Level support). *The total number of integral periodic points on* $S_k(\mathbb{Z})$ *is* 1 *when* $k = 0$, 26 *when* $k = 4$, 6 *at every other positive square level,* 16 *at every other level* $m^2 - m + 2$ *with* $m \geq 1$, *and* 0 *otherwise.*

We prove the classification in Sections 2 and 3; the boundedness statement, Corollary 1.2, and exact all-time return laws follow in Section 4.

1.1 Classical inputs and the scope of the contribution

In the half-trace coordinates $(X, Y, Z) = (x, y, z)/2$, the map is $(X, Y, Z) \mapsto (Y, Z, 2YZ - X)$ and the classical invariant is

$$I = X^2 + Y^2 + Z^2 - 2XYZ - 1 = K/4 - 1. \tag{1.7}$$

Thus our integral lattice becomes the half-integral lattice in that normalization; a change of scale must be tracked in comparing arithmetic statements. Roberts and Baake give the axis cycle, low-period families, and their sign relations in their study of reversible trace maps [4, Eq. (29), Table I, and pp. 849–852]. In particular, the existence of the families in (1.3) and the relation between the four- and twelve-periodic families are not new assertions. We verify their itineraries here because the exact integral classification uses their least periods and signed members.

Roberts develops escape regions and growth criteria for trace maps over the real and complex numbers [3, §§4–5]. Our argument does not improve those quantitative results. Instead, it resolves the arithmetic alternatives at the complement and boundary of possible escape: the equality case at modulus two, the one-zero branch, the signed nonzero branch, and the unit cube. Its principal conclusion is the exhaustiveness of (1.5), not a new construction of its displayed orbits.

Two other nearby questions have different quantifiers. Humphries classifies points with finite orbit for the entire trace-map group arising from $\mathrm{Aut}(F_2)$ [2, Theorem 1]. Periodicity under one particular group element is weaker; the distinction is illustrated in Section 5. Ghosh and Sarnak study the arithmetic of integral points on the surfaces (1.2) and their orbits under the Markoff group [1]. A finite number of group orbits is not the assertion that each such orbit is finite, and it does not enumerate cycles of the fixed word T. Finally, the residual periodicity considered by Vishkautsan concerns a two-reflection map on the zero Markoff surface and reduction modulo primes [5, §§1.1–1.2]; it is not the all-level, single-map integral problem here. In particular, we do not present the zero-level specialization alone as a new contribution.

These comparisons delimit the theorem rather than claim complete literature priority. All arguments establishing Theorem 1.1 are included below and use only the recurrence, integer arithmetic, and finiteness of bounded lattice subsets.

2 Scalar words and exact itineraries

Writing $T^iP = (x_i, x_{i+1}, x_{i+2})$ turns the map into the two-sided scalar recurrence

$$x_{i+3} = x_{i+1}x_{i+2} - x_i, \qquad x_{i-1} + x_{i+2} = x_i x_{i+1}, \quad i \in \mathbb{Z}. \tag{2.1}$$

Three consecutive entries determine every entry in both directions. The inverse in (1.1) follows directly from this recurrence. Also,

$$\begin{aligned} K(T(x, y, z)) &= y^2 + z^2 + (yz - x)^2 - yz(yz - x) \\ &= x^2 + y^2 + z^2 - xyz = K(x, y, z), \end{aligned}$$

which proves the invariance in (1.2).

Lemma 2.1 (Reversal)**.** *If $(x_i)_{i\in\mathbb{Z}}$ satisfies (2.1), then so does $y_i = x_{-i}$. Periodicity and the maximum absolute coordinate are preserved under this reversal.*

Proof. The second equality in (2.1), with index $-i-1$, is $x_{-i-2} + x_{-i+1} = x_{-i-1}x_{-i}$. This is exactly $y_{i-1} + y_{i+2} = y_i y_{i+1}$, with the summands and factors interchanged. The remaining assertions follow from the bijective reindexing of $\mathbb{Z}$. □

Proposition 2.2 (The displayed orbits)**.** *The sets in Table 1 are periodic orbits with exactly the stated least periods, heights, and levels. They are pairwise disjoint, including across all values of $m \geq 1$.*

Proof. Read each of the following scalar words cyclically:

$$A_m : (m, 0, 0, -m, 0, 0), \tag{2.2}$$

$$B_m : (-1, m, -1, 1 - m), \tag{2.3}$$

$$C_m : (1, m, 1, m - 1, -1, -m, 1, 1 - m, 1, -m, -1, m - 1). \tag{2.4}$$

Each consecutive triple is a point, and each cyclic shift is one application of T. Direct substitution in (2.1) verifies the first two words. For example, in the four-letter word the first newly generated entry is $m(-1) - (-1) = 1 - m$, the next is $(-1)(1 - m) - m = -1$, and the remaining two are m

j	$T^j(1,m,1)$	j	$T^j(1,m,1)$
0	$(1,m,1)$	6	$(1,1-m,1)$
1	$(m,1,m-1)$	7	$(1-m,1,-m)$
2	$(1,m-1,-1)$	8	$(1,-m,-1)$
3	$(m-1,-1,-m)$	9	$(-m,-1,m-1)$
4	$(-1,-m,1)$	10	$(-1,m-1,1)$
5	$(-m,1,1-m)$	11	$(m-1,1,m)$

Table 2: The twelve consecutive triples of C_m. The numbering refers to ordinary iterates of T, with no sign quotient or change of clock.

and -1. For the twelve-letter word, the complete list of triples is shown in Table 2; applying T to index j gives index $j+1$ modulo twelve.

The positive value m occurs once in (2.2). Therefore that scalar word cannot be a repetition of a shorter word, and the least period is six. For $m \geq 2$ the same unique occurrence of the positive value m in (2.3) and (2.4) proves least periods four and twelve. This argument needs a separate check at $m = 1$, where the value 1 occurs more than once. In that case the four triples of B_1 are

$$(-1,1,-1), \quad (1,-1,0), \quad (-1,0,-1), \quad (0,-1,1), \tag{2.5}$$

which are distinct. The twelve triples of Table 2 at $m = 1$ are likewise pairwise distinct: those containing a zero have different ordered positions or signs, and the four without a zero are $(1,1,1), (-1,-1,1), (1,-1,-1), (-1,1,-1)$. Hence the stated least periods hold at this endpoint as well.

For the four points in $E \sqcup D$, each coordinate has modulus two and their product is positive. Thus $yz = 2x$ at each point, so that $T(x,y,z) = (y,z,x)$. This gives the fixed point E and the three-cycle D. The origin is also fixed.

Since $|m-1| \leq m$ for $m \geq 1$, the scalar words have heights exactly m. At the representatives, direct evaluation gives

$$K(m,0,0) = m^2, \qquad K(-1,m,-1) = K(1,m,1) = m^2 - m + 2. \tag{2.6}$$

The special levels follow by substitution. The three infinite families have distinct least periods, so no orbit belongs to two different families. Within a family, different m give different heights. The special orbits have least periods one or three and are disjoint from those families; O and E are different points. This proves all the assertions. $\square$

The proposition establishes inclusion of the right-hand side of (1.5) in the periodic set. It does not yet exclude other periods. That exclusion is the next section's purpose.

3 Exhaustiveness by a maximal coordinate

Let (x_i) be a periodic integral solution of (2.1), and set $M = \max_i |x_i|$. If $M = 0$, the point is the origin. We first handle every $M \geq 2$ uniformly, and then resolve $M = 1$ by a complete count within the unit cube.

Choose the index so that $x_0 = u$ with $|u| = M$, and write

$$a = x_{-1}, \qquad b = x_1. \tag{3.1}$$

Two instances of the recurrence give

$$au = x_{-2} + b, \qquad ub = a + x_2. \tag{3.2}$$

No assumption has been made on the sign of u.

Lemma 3.1 (Neighbour bound and equality case)**.** *For $M \geq 2$, the neighbours satisfy $|a|, |b| \leq 2$. If either has modulus two, the orbit belongs to $E \sqcup D$. Every other periodic orbit of height at least two therefore has $a, b \in \{-1, 0, 1\}$ at a coordinate of maximal modulus.*

Proof. The triangle inequality in (3.2) gives $M|a| \leq 2M$ and $M|b| \leq 2M$, proving the bound. Suppose first that $b = 2\eta$ with $\eta \in \{-1, 1\}$. The equality $2\eta u = a + x_2$, together with $|a|, |x_2| \leq M = |u|$, forces equality in the triangle inequality. Both terms have the sign of ηu and modulus M:

$$a = x_2 = \eta u. \tag{3.3}$$

The first equation in (3.2) now implies

$$M^2 = |au| \leq |x_{-2}| + |b| \leq M + 2. \tag{3.4}$$

Since M is an integer at least two, this forces $M = 2$. Writing $u = 2\varepsilon$, the consecutive triple at the maximum is

$$(a, u, b) = (2\eta\varepsilon, 2\varepsilon, 2\eta), \qquad \varepsilon, \eta \in \{-1, 1\}. \tag{3.5}$$

Its entries have modulus two and positive product. The four choices are exactly $E \sqcup D$, already verified in Proposition 2.2. Three consecutive entries determine the entire two-sided orbit, so there is no further continuation in this case.

If $|a| = 2$, apply the preceding argument to the reversed scalar sequence of Lemma 2.1. Its right neighbour is a. The resulting triple is a permutation of a point of $E \sqcup D$, and hence again lies in that same four-point set. In all other cases, integrality and the neighbour bound give the final assertion. □

Lemma 3.2 (Zero-neighbour alternatives)**.** *Suppose $M \geq 2$ and $a, b \in \{-1, 0, 1\}$. If both neighbours vanish, the orbit is A_M. Exactly one vanishing neighbour is impossible.*

Proof. If $a = b = 0$, the triple is $(0, u, 0)$, with $u = \pm M$. Both signs occur in the already verified axis word (2.2), so uniqueness of the scalar sequence gives A_M.

Suppose that $a = 0$ and $b = t \in \{-1, 1\}$. Starting from $(x_{-1}, x_0, x_1) = (0, u, t)$, the recurrence gives

$$\begin{aligned} &x_2 = ut, && x_3 = 0, && x_4 = -t, \\ &x_5 = -ut, && x_6 = u, && x_7 = t(1 - u^2). \end{aligned} \tag{3.6}$$

In the final step we used $t^2 = 1$. It follows that $|x_7| = M^2 - 1 > M$ for every integer $M \geq 2$, contradicting the definition of M. This includes the endpoint $M = 2$. The case $b = 0$, $a \in \{-1, 1\}$ reduces to the one just treated by Lemma 2.1. □

Lemma 3.3 (Nonzero-neighbour alternatives)**.** *Suppose $M \geq 2$ and $a, b \in \{-1, 1\}$. Then the orbit is B_M or C_M.*

(a, b)	(a, u, b)	Identified point
$(-1, -1)$	$(-1, M, -1)$	The representative of B_M
$(1, 1)$	$(1, M, 1)$	The representative of C_M
$(-1, 1)$	$(-1, -M, 1)$	$T^4(1, M, 1)$ in C_M
$(1, -1)$	$(1, -M, -1)$	$T^8(1, M, 1)$ in C_M

Table 3: All signed nonzero-neighbour possibilities at the maximum. The two nontrivial time labels are read directly from Table 2.

Proof. The second equation of (3.2) reads $x_2 = bu - a$. If $abu < 0$, the terms bu and a have opposite signs, so $|x_2| = M + 1$, a contradiction. Thus $abu > 0$, and therefore $u = abM$. This leaves precisely the four triples in Table 3.

Every row is an already verified point of the stated orbit. Uniqueness of the recurrence completes the proof. In particular the argument has not merged the four-cycle and twelve-cycle by passing to a sign quotient. ☐

Proposition 3.4 (The unit-cube remainder)**.** *Every periodic orbit of height at most one belongs to* $O \sqcup A_1 \sqcup B_1 \sqcup C_1$.

Proof. Consider triples in $\{-1, 0, 1\}^3$. If a triple has no zero entry and negative coordinate product, then $yz = -x$ and the next coordinate is $yz - x = -2x$, of modulus two. The four triples of this type cannot occur in an orbit of height at most one.

The remaining 23 triples consist of the origin, the six triples with exactly one nonzero entry, the twelve with exactly two nonzero entries, and the four with three nonzero entries and positive product. The first two groups are O and A_1. Equations (2.5) and (2.4), or Table 2, show that B_1 and C_1 lie in the union of the last two groups. They are disjoint and have total size $4 + 12 = 16$, by Proposition 2.2. That union has exactly sixteen elements, so they exhaust it. This proves the claim without a search or an assumed numerical period bound. ☐

Proof of the periodic classification in Theorem 1.1. If $M \geq 2$, Lemma 3.1 either identifies the orbit as E or D, or reduces its neighbours to $\{-1, 0, 1\}$. Lemmas 3.2 and 3.3 then identify it as A_M, B_M, or C_M. Proposition 3.4 handles $M \leq 1$. This proves inclusion in the right-hand side of (1.5); the reverse inclusion, disjointness, and least periods were proved in Proposition 2.2. ☐

As a finite consequence, the cube $[-2, 2]^3$ contains exactly 49 points whose entire orbit remains in that cube: the five special points and the 22 points contributed by each of $m = 1, 2$. For every other one of its 76 initial states, each time direction must leave the cube in at most 125 iterates. Indeed, 126 successive states inside a set of size 125 would repeat; invertibility would then make the initial point periodic. The classification just proved would force it into the listed 49 points. This finite certificate is a consequence of the uniform proof, not an input to it.

4 Proper escape and arithmetic return laws

4.1 Bounded half-orbits and two-sided escape

The passage from periodic classification to integral escape uses discreteness rather than a real dynamical estimate.

Lemma 4.1 (A discrete-bijection principle)**.** *Let F be a bijection of a discrete subset X of a finite-dimensional real vector space such that every bounded subset of X is finite. A point of X has a bounded forward or backward orbit if and only if it is periodic. Every nonperiodic two-sided orbit visits each bounded subset only finitely many times.*

Proof. A bounded forward orbit lies in a finite set, so $F^iP = F^jP$ for some $0 \le i < j$. Applying F^{-i} gives $P = F^{j-i}P$. The same argument with F^{-1} treats backward orbits. Conversely, a periodic orbit is finite and bounded.

For a nonperiodic point, the points F^nP, $n \in \mathbb{Z}$, are pairwise distinct: any equality would give a period by applying an inverse iterate. A finite subset can contain only finitely many of these distinct points. This proves the last assertion. □

Apply the lemma to $X = \mathbb{Z}^3$ and $F = T$. For any finite radius R, only finitely many integers n have $T^nP \in [-R, R]^3$ when P is nonperiodic. Thus, beyond the last visit in either time direction, $\|T^nP\|_\infty > R$. Since R is arbitrary, this proves (1.6) and completes the boundedness part of Theorem 1.1.

4.2 Every level and every ordinary time

For $k \in \mathbb{Z}$, define

$$s_k = \mathbf{1}_{\{k=m^2 \text{ for some } m\ge1\}}, \qquad q_k = \mathbf{1}_{\{k=m^2-m+2 \text{ for some } m\ge1\}}. \tag{4.1}$$

Both parametrizations are injective on positive integers. The second condition is equivalently that $4k-7$ be a positive odd square, because $4(m^2-m+2)-7 = (2m-1)^2$. Let $c_d(k)$ denote the number of orbits of least period d on $S_k(\mathbb{Z})$. The classification immediately gives

$$\begin{aligned} &c_1(k) = \mathbf{1}_{k=0} + \mathbf{1}_{k=4}, && c_3(k) = \mathbf{1}_{k=4}, && \\ &c_4(k) = q_k, && c_6(k) = s_k, && c_{12}(k) = q_k, \end{aligned} \tag{4.2}$$

and $c_d(k) = 0$ for every other d. In particular these counts are finite on each fixed level, whether or not that level contains infinitely many nonperiodic integral points.

Corollary 4.2 (All-time fixed-point counts)**.** *For every $k \in \mathbb{Z}$ and $n \ge 1$,*

$$\begin{aligned} F_k(n) := \#\,\mathrm{Fix}(T^n; S_k(\mathbb{Z})) = \mathbf{1}_{k=0} &+ \mathbf{1}_{k=4}(1 + 3\mathbf{1}_{3|n}) \\ &+ 6s_k\mathbf{1}_{6|n} + q_k(4\mathbf{1}_{4|n} + 12\mathbf{1}_{12|n}). \end{aligned} \tag{4.3}$$

Proof. An orbit of least period d contributes all d of its points to $\mathrm{Fix}(T^n)$ if $d \mid n$, and none otherwise. Therefore $F_k(n) = \sum_{d|n} dc_d(k)$, and (4.2) gives the result. □

Definition 4.3 (Fixed-level dynamical zeta function)**.** The ordinary dynamical zeta function of the integral points on a fixed level is the formal power series

$$\zeta_k(t) = \exp\left(\sum_{n\ge1} \frac{F_k(n)}{n} t^n\right) \in 1 + t\mathbb{Q}[[t]]. \tag{4.4}$$

Corollary 4.4 (Exact fixed-level zeta function)**.** *For every integer k,*

$$\zeta_k(t) = \frac{1}{(1-t)^{\mathbf{1}_{k=0}+\mathbf{1}_{k=4}}(1-t^3)^{\mathbf{1}_{k=4}}(1-t^6)^{s_k}(1-t^4)^{q_k}(1-t^{12})^{q_k}}. \tag{4.5}$$

This is also an analytic identity for $|t| < 1$ and gives a rational continuation to the complex plane.

Proof. For a single orbit of length d, its contribution to the exponent of (4.4) is

$$\sum_{r\geq 1}\frac{d}{dr}t^{dr} = -\log(1-t^d).$$

Only finitely many orbits occur at any fixed k, so summing these contributions gives $\zeta_k(t) = \prod_d (1-t^d)^{-c_d(k)}$, which is (4.5). Formula (4.3) bounds $F_k(n)$ uniformly in n, so the defining logarithmic series is absolutely convergent for $|t|<1$ as well. $\square$

This product is over source periodic orbits. Its index is a period, not a rational prime or a residue-field degree. The rationality in (4.5) follows from the finite periodic locus on the fixed level, not from an additional arithmetic correspondence.

4.3 The intersection of the two supports

Lemma 4.5. *The sets* $\{r^2 : r\geq 1\}$ *and* $\{m^2-m+2 : m\geq 1\}$ *intersect only at* 4.

Proof. Suppose $r^2 = m^2-m+2$ with $r, m\geq 1$. Completing the square and factoring give

$$(2r-2m+1)(2r+2m-1) = 7. \tag{4.6}$$

Since $r^2 = (m-\frac{1}{2})^2+\frac{7}{4}$, we have $r > m-\frac{1}{2}$. Both factors in (4.6) are therefore positive integers. The second is larger than the first because their difference is $4m-2>0$. As 7 is prime, they must be 1 and 7. Adding the two equations gives $r=2$, and subtracting gives $m=2$. Conversely those values give level 4. $\square$

The total periodic-point count is

$$\sum_{d\geq 1} dc_d(k) = \mathbf{1}_{k=0} + 4\mathbf{1}_{k=4} + 6s_k + 16q_k. \tag{4.7}$$

Lemma 4.5 makes its alternatives disjoint, as recorded in Table 4; this proves Corollary 1.2.

Level condition	Periodic orbits	Number of points
$k=0$	O	1
$k=4$	E, D, A_2, B_2, C_2	26
$k=m^2\neq 4$, $m\geq 1$	A_m	6
$k=m^2-m+2\neq 4$, $m\geq 1$	B_m, C_m	16
All remaining $k\in\mathbb{Z}$	None	0

Table 4: The complete periodic-point count on every integer level. The alternatives are mutually exclusive by Lemma 4.5.

There is no ordinary finite-count zeta function on the unrestricted lattice obtained by removing k from the notation. For example, every point of A_m is fixed by T^6, and the disjoint axis orbits exist for all $m\geq 1$. Hence $\#\,\mathrm{Fix}(T^6;\mathbb{Z}^3)=\infty$. The fixed-level quantifier in (4.4) is indispensable.

5 Scope, verification, and conclusion

5.1 Why the arithmetic domain matters

Theorem 1.1 is not a classification of rational periodic points. A concrete rational two-cycle is

$$T(3, 3/2, 3) = (3/2, 3, 3/2), \qquad T(3/2, 3, 3/2) = (3, 3/2, 3). \tag{5.1}$$

The two points are distinct, so the least period is two, which is excluded over $\mathbb{Z}$ by the theorem. The neighbour bound of Lemma 3.1 alone would not reduce rational neighbours to three integer values. Thus even a rational periodic point on a real invariant level cannot be inserted into the integral proof.

Whole-group finiteness implies periodicity under each individual group element, but the converse implication fails for a single chosen map. For example, $(-1, 3, -1) \in B_3$ has $K = 8$, is not on an axis, and has least period four under T. In half-trace coordinates it is $(-1/2, 3/2, -1/2)$, lying on $X^2 + Y^2 + Z^2 - 2XYZ = 2$. It is excluded from the non-axis finite-orbit alternatives in Humphries's whole-group theorem [2, Theorem 1 and its proof]. That theorem explicitly treats axes separately; the distinction here does not rely on incorrectly identifying all finite group orbits with characters of finite matrix groups.

5.2 What the exact checks do and do not show

The proof of exhaustiveness is the symbolic argument in Sections 2 and 3. No executable calculation, height cutoff, or bound on a possible period is needed to establish its universal quantifiers. Supplementary exact-integer checks were nevertheless used during verification to test the signed itineraries and finite consequences.

The recorded checks verify the polynomial identities for the displayed families and the rational example, reconstruct the finite orbit graph on the 68,921 states of $[-20, 20]^3 \cap \mathbb{Z}^3$, and check fixed-level return formulas for $-10 \le k \le 200$ and $1 \le n \le 60$. The finite graph has 445 states on cycles wholly contained in its cube: two fixed points, one three-cycle, and twenty cycles of each of lengths four, six, and twelve. On the smaller cube $[-2, 2]^3$, the checks recover 49 states on cycles wholly contained in the cube and verify that each of the other 76 initial states leaves this cube in each time direction within nine iterates. Leaving this cube does not imply global nonperiodicity: for example, $(-1, -2, -1) \in B_3$ lies in the cube but maps to $(-2, -1, 3)$ outside it. This last observed bound is stronger than the elementary 125-step certificate above, but neither bound is used to infer anything about arbitrary heights. The exact check record and source-reading record accompany the manuscript; they supplement, and do not replace, the full proof or the distinction between mathematical correctness and literature priority.

5.3 Conclusion

The classical axis and low-period families exhaust the integral periodic locus of the single Fibonacci trace map on all invariant levels. The proof reduces an unbounded family of possible integer cycles to a uniform local maximum argument, with all equality, zero, sign, and small-height cases resolved. Discreteness then gives two-sided proper escape for every other integer point, and the classification determines the ordinary return counts and rational zeta function of each fixed level.

These conclusions do not furnish a classification over $\mathbb{Q}$, a new real or complex escape rate, or finite-field point counts. The source-orbit factors in (4.5) do not establish target Euler factors, root numbers, a zero or divisor correspondence, or a Hilbert–Pólya realization. Those are additional questions requiring objects and comparison theorems not asserted here.

References

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